FASTSPEECH 2

English and Telugu Technical Report

Implementation, training results and traceable variance analysis

Evidence reviewed: 2 November 2026. Separate acoustic models; one frozen Universal HiFi-GAN vocoder.

Selected system	English	Telugu
Checkpoint	12000	8000
Training / validation clips	3135 / 626	1877 / 357
Best validation total loss	2.6313	2.0653
Final-sentence listening	Complete; partly noisy	Complete; clear
Conditioning	115 demographic proxies	2 known speaker groups

Summary

The project implements separate non-autoregressive FastSpeech 2 acoustic models for English and Telugu, using the pinned reference architecture with compatibility and preprocessing patches. Both predict mel spectrograms, duration, pitch and energy; a pretrained Universal HiFi-GAN converts predicted mels to waveforms. Final inference uses only phoneme tokens and the chosen conditioning ID, without reference acoustic targets.

The owner reported all five English test sentences complete but partly noisy, and all five Telugu test sentences clear and complete. These observations are subjective listening results. No MOS, ASR score, word-level accuracy or numerical perceptual improvement was measured. Numerical checks and readable plots do not establish pronunciation or boundary accuracy.

Evidence and review boundary

This report combines the supplied results bundle, reviewed source/configuration bundle, supplementary token and inference metadata, and recorded Colab execution/listening observations. The review environment checked saved arrays, hashes recorded in metadata, normalization correspondence, syntax and rendered figures. It did not load acoustic-model weights, rerun GPU training, inspect the final WAV bytes or listen to them independently.

The selected English system is the original model. The separate data_repair_v1 run is not the submitted English model. Publishing and clean-environment reproduction remain integration tasks; this package does not include external weights or audio.

1. Data and split construction

Final successful features	English	Telugu
Training clips	3135	1877
Validation clips	626	357
Training source hours	4.99613	4.97338
Validation source hours	0.99647	0.97204
Training aligned feature hours	4.58661	4.39903
Validation aligned feature hours	0.90453	0.85925

Source duration includes the selected recordings; aligned feature duration is shorter after trimming boundary silence and frame conversion. Durations are from the recorded extraction/audit outputs, not a new decoding pass in this review.

English: Svarah

The recorded source audit found 6656 English utterances, about 9.61444 hours, at 16 kHz. The dataset revision was ebbf7777fe771490696a3f7b007097606fa8c924. The primary-language metadata describes speakers' native languages; it does not turn these recordings into Telugu speech. Repeated normalized text and recording-group keys were grouped during split construction. Seed 42 and a 1-20 second duration window were used.

No verified per-recording speaker identity was available in the chosen metadata. The implementation uses 115 demographic proxy groups, not 115 verified individual speakers. Validation proxy coverage was checked, but the experiment must not be described as speaker-disjoint evaluation or reliable single-person voice cloning.

Telugu: IndicTTS_Telugu

The recorded audit found 8576 clips totaling about 36.71227 hours, rather than relying on the advertised dataset duration. The revision was 7ab3c18491bef3f5d4e820c2701b5f78f54031d6. Two conditioning groups represent female and male recordings. Audited female files were 16 kHz and male files 48 kHz. Selection balanced the two groups and grouped identical transcripts across splits. Validation uses known speaker groups.

Exclusions and review limits

Telugu training mismatches 191, 913 and 2652 and validation mismatches 915, 920 and 2646 were excluded after owner listening. A validation candidate with unsupported training phones was deferred. English alignment/transcript review and pitch-extraction failures also reduced the original selections to the final counts. Structural checks passed, but full-corpus listening and phone-boundary review were not completed.

A later English repair candidate excluded mismatch 1346 and deferred alignments 4240 and 6298, leaving 3132 training clips. Sixteen speaking-rate flags remained unreviewed at the owner's request. Only a separate 1000-step repair run was reported; no perceptual improvement was established and its statistics were not used for the final English selection.

2. Architecture and acoustic targets

Setting	Implementation
Encoder / decoder	4 / 6 FFT blocks; 2 attention heads; hidden width 256
FFT convolution	Filter size 1024; kernel sizes 9 and 1; dropout 0.2
Variance predictors	Filter size 256; kernel size 3; dropout 0.5
Variance embeddings	Linear pitch/energy quantization; 256 bins
Mel / waveform	80 channels; 22050 Hz waveform
STFT / mel band	FFT/window 1024; hop 256; mel limits 0-8000 Hz
Model parameters	English 35,188,801; Telugu 35,171,393
Symbols / conditioning	English 360 / 115 proxies; Telugu 405 / 2 groups

The saved model configuration uses learned conditioning embeddings. Telugu adds 45 training-supported phone aliases to the reference symbol inventory in a separate repository. The architecture predicts phoneme-level duration, pitch and energy, expands hidden states to frames, then predicts mel features with a PostNet refinement. This implementation uses direct phoneme-level pitch targets rather than a wavelet pitch representation.

Text and alignment

English uses ARPAbet candidates from a probability-stripped synthesis dictionary, with G2P fallback. Alignment and synthesis dictionaries are kept separate. Telugu uses wordwise eSpeak NG 1.51 IPA candidates, removes stress marks and maps 45 supported phones to TEAA-TEBS aliases. Unsupported phones are rejected, not silently dropped. Candidate pronunciation coverage is not pronunciation accuracy.

English alignments used the pretrained english_us_arpa acoustic model. Telugu required a separate train-only MFA aligner: monophone 40 iterations/1000 Gaussians; triphone and SAT each 35 iterations, 1000 leaves/5000 Gaussians. MFA features were 16 kHz MFCCs at 10 ms. Difficult training alignments were retried; no validation audio trained the Telugu acoustic aligner.

Audio and feature extraction

Alignment audio was standardized to 16 kHz. Fourteen Telugu resampling overshoots received a recorded uniform gain to peak 0.999 rather than sample clipping. Original shard audio remained unchanged. TTS features were then extracted at 22050 Hz from these exports; upsampling cannot recover bandwidth lost at 16 kHz. WORLD-based pitch was interpolated through unvoiced regions, then duration-averaged to phones. Energy uses the extractor's STFT scale.

3. Normalization and implementation checks

Training-only fit	English pitch	English energy	Telugu pitch	Telugu energy
Mean	180.656532	28.538314	196.214109	32.261455
Standard deviation	60.012250	26.858434	45.438429	23.983964
Positive-duration observations	161097	161097	192141	192141

Normalized value = (raw value - training mean) / training standard deviation. Zero-duration positions are excluded from fitting and set to normalized zero. English has 113 such training positions and 19 validation positions; Telugu has 94 and 10. No outlier filtering was applied. Embedding-bin bounds are fitted on training values only; validation is not clipped.

The supplementary stats.json files exactly reproduce the saved inverse-normalized pitch and energy arrays for all ten final sentences within numerical tolerance. Frozen token IDs and original text also match their generated JSON records. Every final duration array is finite, nonnegative, integer-valued and sums to its saved mel-frame count. No zero-duration final phones were found.

Compatibility changes

The shared audio patches update librosa keyword-only calls. The extractor sets load rate explicitly, checks feature lengths, preserves empty TextGrid intervals and maps empty interval labels to sp. This prevents omitted internal gaps from corrupting duration correspondence. Language-specific loaders use CPU map_location and weights_only checkpoint loading; the scheduler stores learning rates as Python floats. Telugu alone receives the appended phone symbols.

Vocoder and checkpoint checks

The official Universal HiFi-GAN generator is frozen and was not fine-tuned. Its configuration matches the model mel settings. Owner listening found original recordings and real-mel reconstructions clear. Selected model checkpoints were reported finite with correct Adam update counts. In this review, selected checkpoint hashes in all prediction records match the saved selection; the weight files themselves were not uploaded or rehashed here.

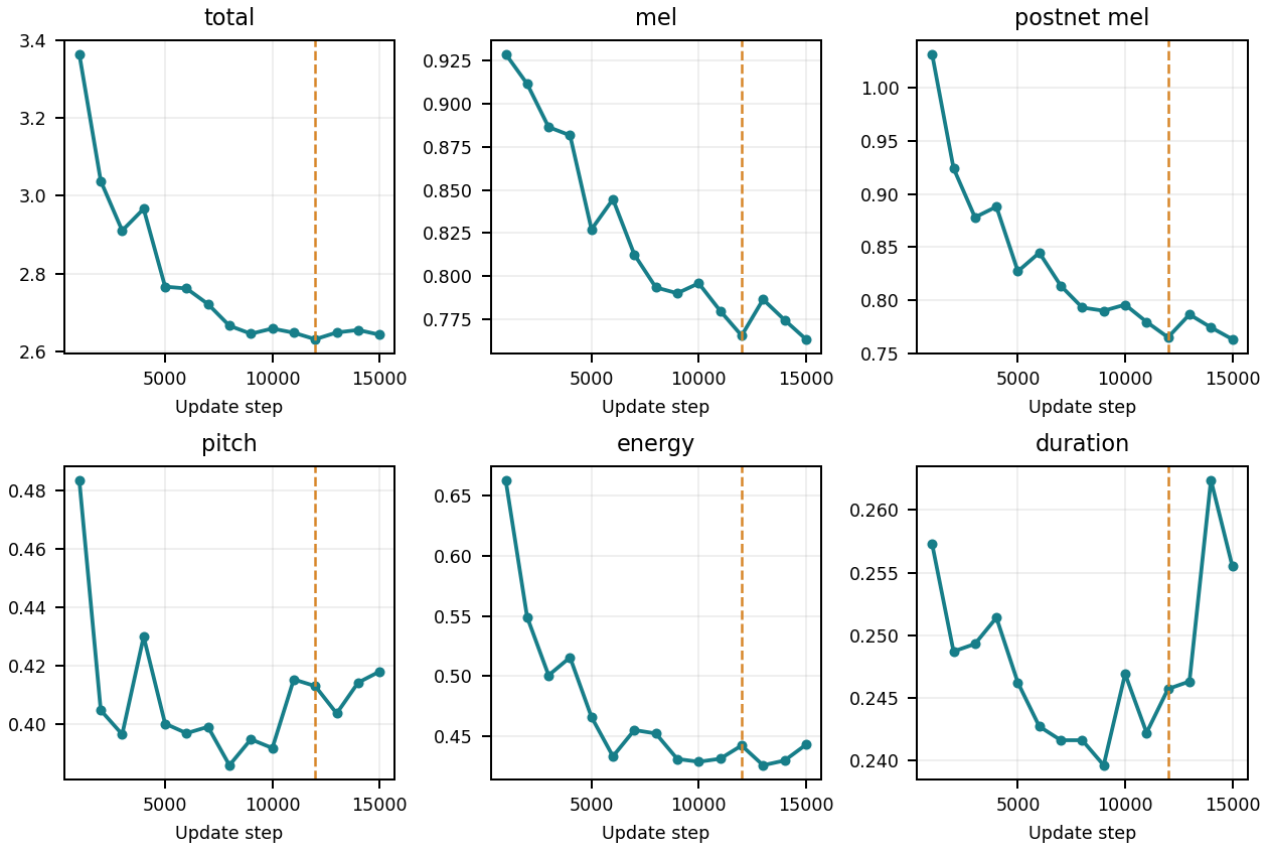
The inference scripts check predicted waveform finiteness and peak range before PCM-16 export. Prior diagnostics found no range overflow in the noisy English examples. This weakens a simple clipping explanation but does not prove an acoustic cause. Both pre- and post-PostNet English examples were reported noisy, so disabling the PostNet was not established as a repair.

Runtime and resumptions

Training used a Tesla T4, batch size 4, gradient accumulation 1, Adam betas 0.9/0.98, epsilon 1e-9, warm-up 4000 and gradient clipping 1.0. Checkpoint/validation intervals were 1000 steps. Wrappers reseed with 42 on resume and restore Adam/model state, but not full RNG or data-loader position. A fresh-run bitwise reproducibility claim would therefore be unsupported.

4. Training results: English

Selected validation step	Total	Mel	PostNet mel	Pitch	Energy	Duration
12000	2.6313	0.7654	0.7651	0.4129	0.4422	0.2457



Validation losses from the supplied CSV. Dashed line marks the selected minimum-total-loss checkpoint. Components have different scales; the panels do not measure perceptual quality.

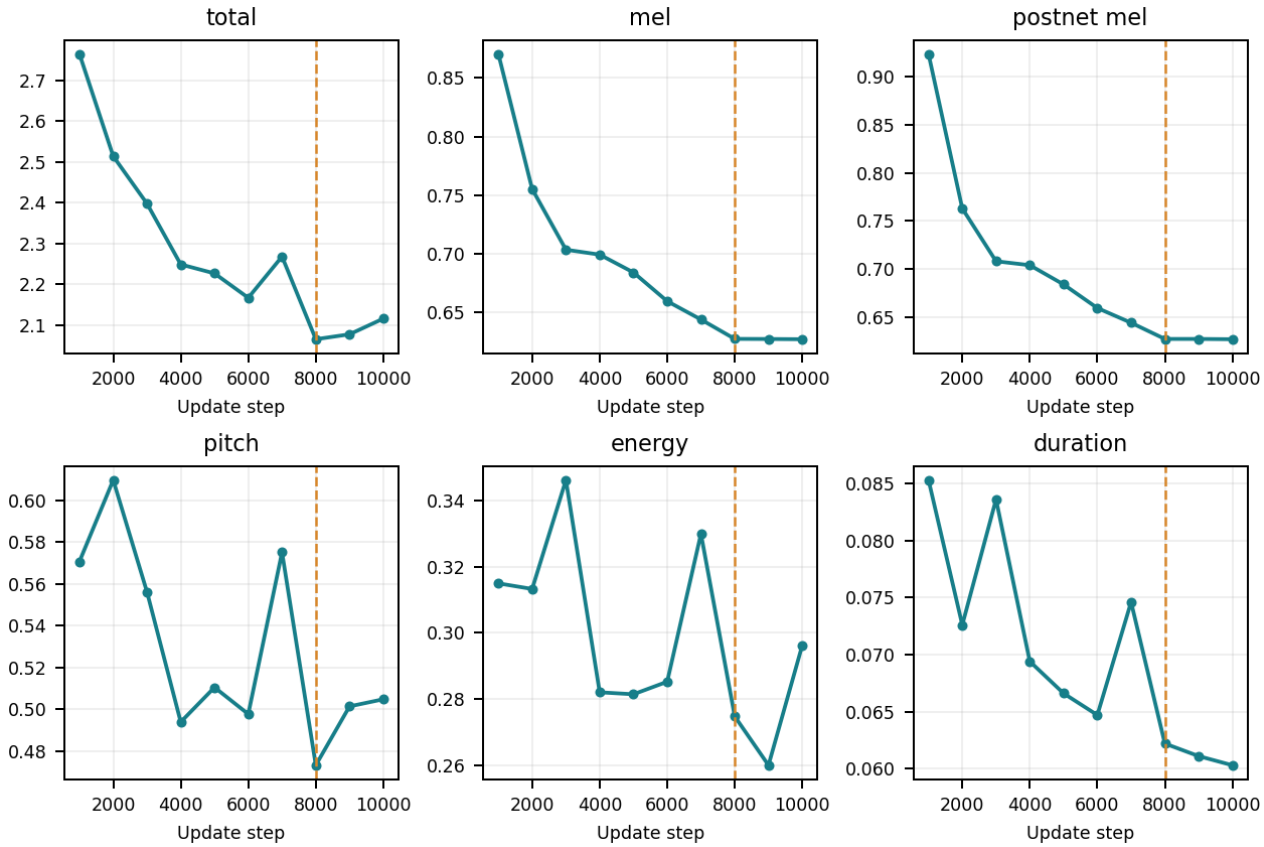
The original English model trained through 15000 updates. The CSV contains 150 logged training minibatches and 15 full validation evaluations. Total validation loss decreases from 3.3646 at 1000 to 2.6313 at 12000, then remains near 2.64-2.66. At 15000 it is 2.6429. The late curve suggests a plateau, not a reliable further quality gain.

History reconciliation retains the interrupted 5000-start run only through checkpoint 6000, then uses the successful 6000-start run for later points. This avoids combining an abandoned 6100-6900 trajectory with the resumed one. Training curves show sampled minibatch loss, not an epoch mean, so their gap to validation is not by itself a quantified overfitting result.

The owner reported 9000-step outputs noisy and 12000-step outputs partly noisy. All five final English sentences remained partly noisy. Selection at 12000 is based on validation loss plus those observations; it is not a claim of clean English speech.

5. Training results: Telugu

Selected validation step	Total	Mel	PostNet mel	Pitch	Energy	Duration
8000	2.0653	0.6276	0.6275	0.4732	0.2747	0.0622



Validation losses from the supplied CSV. Dashed line marks the selected minimum-total-loss checkpoint. Components have different scales; the panels do not measure perceptual quality.

Telugu trained through 10000 updates, with 100 logged training minibatches and 10 validation evaluations. Total validation loss decreases from 2.7648 at 1000 to 2.0653 at 8000, followed by 2.0774 at 9000 and 2.1158 at 10000. The 8000 checkpoint is the best observed total-loss checkpoint.

The small late increase is consistent with a plateau or emerging overfitting, but two later validation points do not establish the cause. Same-sentence male/female listening at 5000 and 8000 was clear and complete, with no noticeable difference reported. Final assignment generation uses female ID 0 and was reported clear and complete on all five texts.

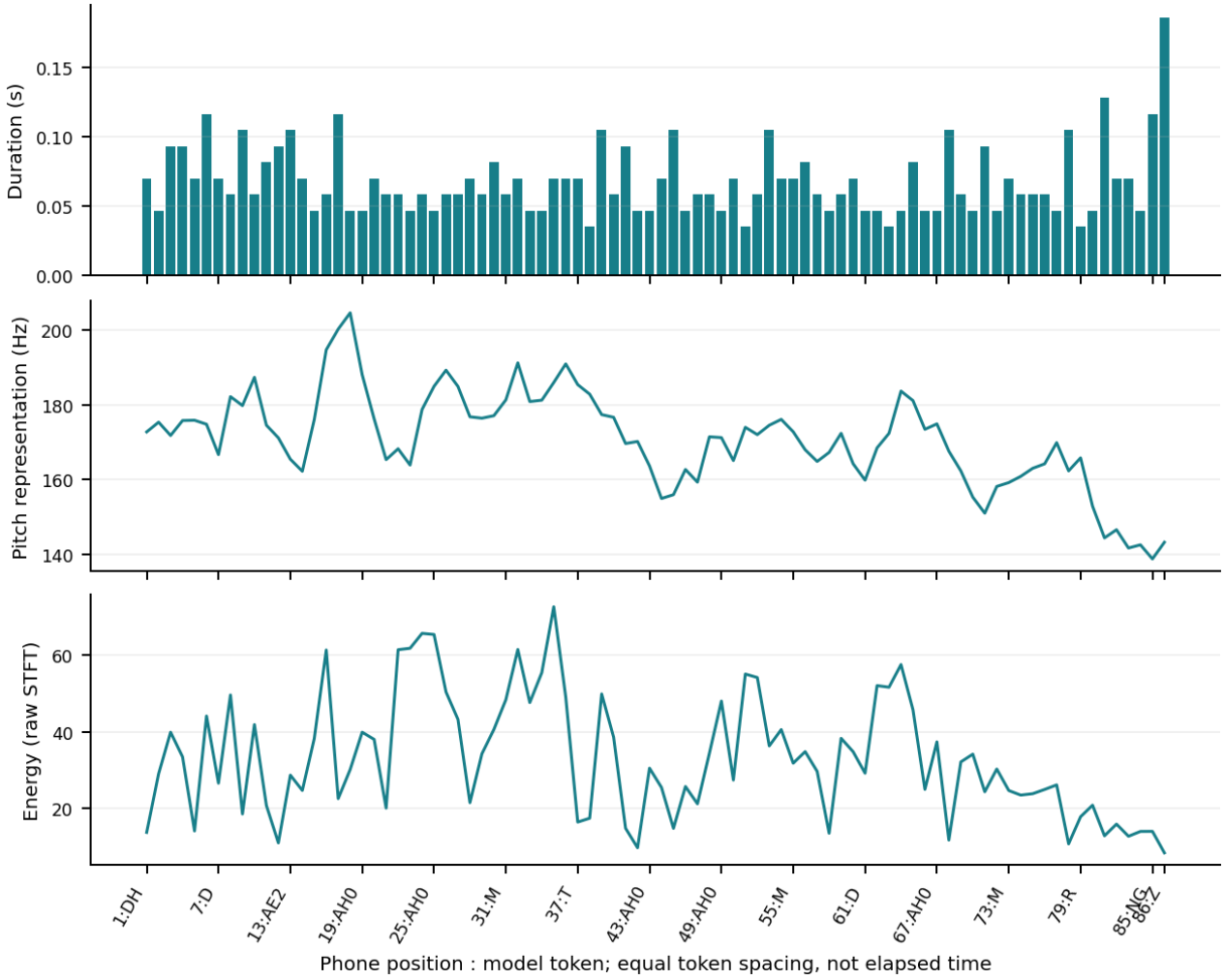
English and Telugu losses should not be ranked as a perceptual language comparison: the datasets, conditioning groups, phoneme inventories and target distributions differ.

E1 | English sentence analysis

Original: The antidisestablishmentarianism movement gained considerable momentum during the parliamentary proceedings.

Selected checkpoint: 12000 | Conditioning: proxy_086 (ID 86) | Phones: 86 | Waveform duration: 5.805 s

Owner listening: partly noisy; complete. Pitch range: 138.8-204.5 Hz representation. Energy range: 8.47-72.55, raw STFT scale.



The long technical word increases sequence length. The early pitch rise and later decline coexist with a relatively long final Z. Token coverage confirms encoding only; the owner did not score pronunciation of antidisestablishmentarianism separately.

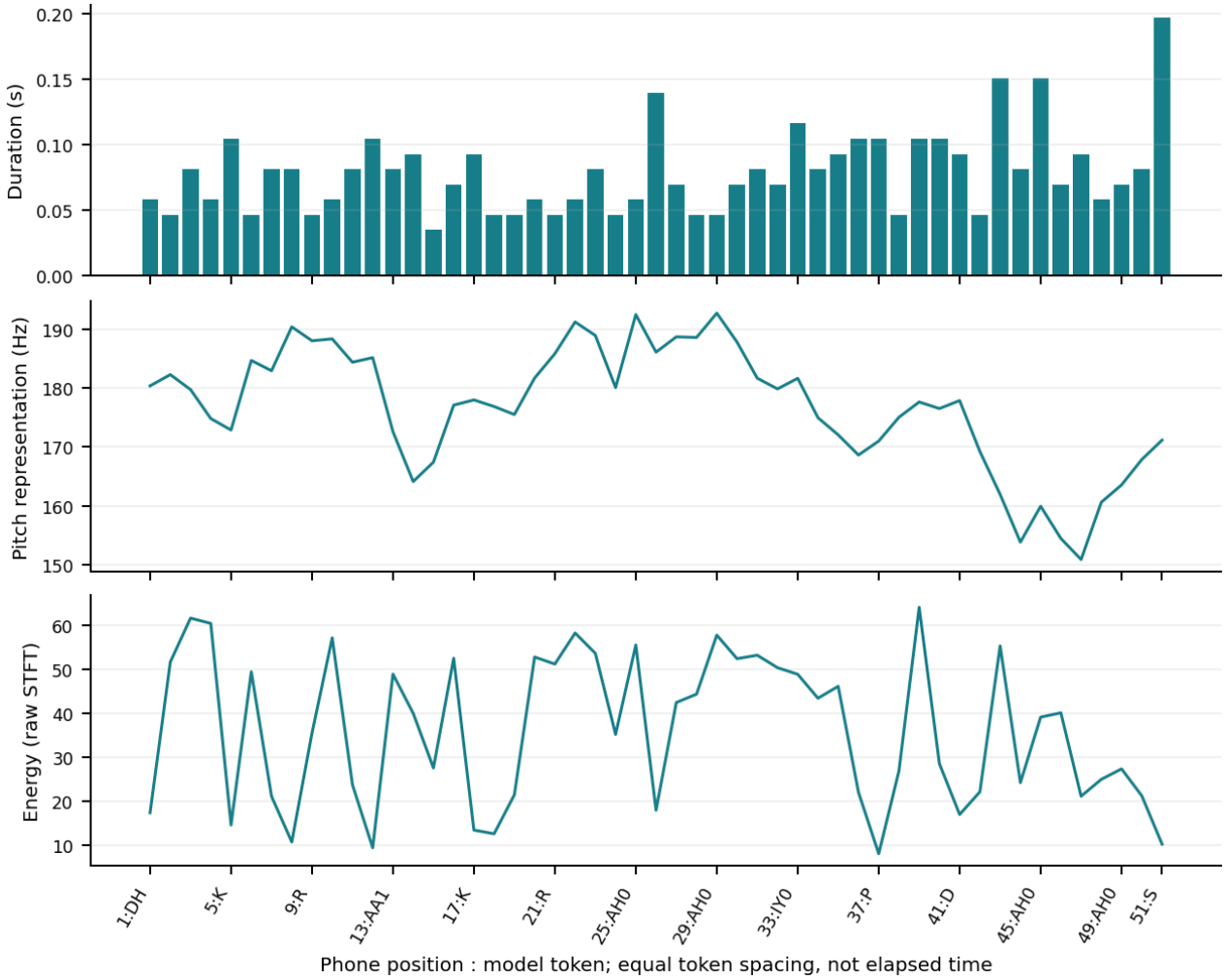
Curves show saved model predictions, not forced-alignment targets or features measured from the generated WAV. Phoneme-level pitch was trained with interpolation through unvoiced regions. There are no reference test prosody targets for an accuracy comparison.

E2a | English sentence analysis

Original: The orchestra's conduct during the ceremony was praised by the audience.

Selected checkpoint: 12000 | Conditioning: proxy_086 (ID 86) | Phones: 51 | Waveform duration: 4.029 s

Owner listening: partly noisy; complete. Pitch range: 150.9-192.7 Hz representation. Energy range: 8.13-64.09, raw STFT scale.



The noun pronunciation of conduct uses K AA1 N D AH0 K T from the dictionary. The final S is relatively long. The sentence was complete but partly noisy; correct lexical stress was not independently scored.

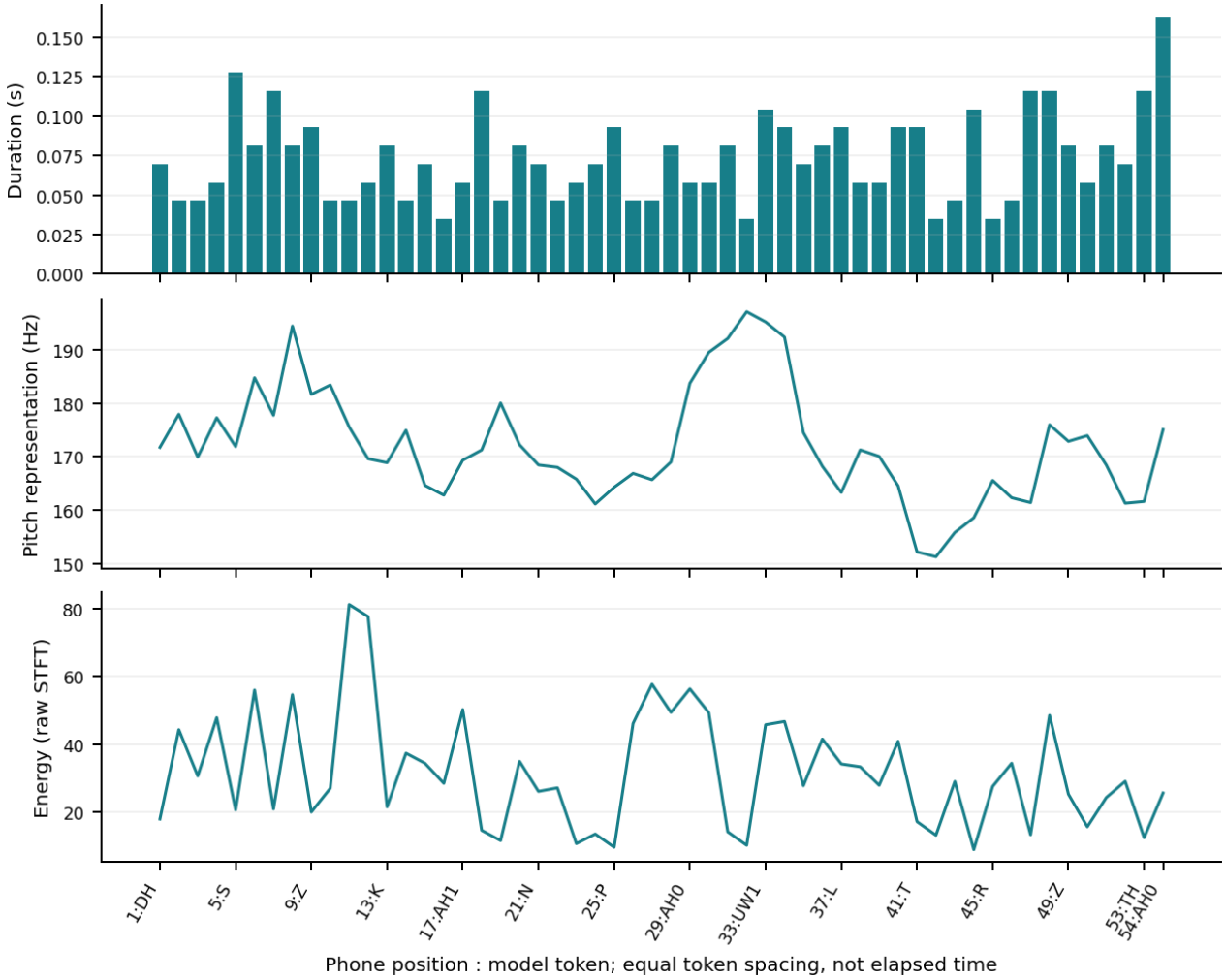
Curves show saved model predictions, not forced-alignment targets or features measured from the generated WAV. Phoneme-level pitch was trained with interpolation through unvoiced regions. There are no reference test prosody targets for an accuracy comparison.

E2b | English sentence analysis

Original: The researchers will conduct an experiment to evaluate the proposed method.

Selected checkpoint: 12000 | Conditioning: proxy_086 (ID 86) | Phones: 54 | Waveform duration: 3.959 s

Owner listening: partly noisy; complete. Pitch range: 151.3-197.1 Hz representation. Energy range: 9.02-81.19, raw STFT scale.



The verb pronunciation of conduct is an explicit frontend override: K AH0 N D AH1 K T. This is not evidence of model-learned heteronym disambiguation. The saved method phones end M EH1 TH AH0, without final D; this remains a frontend limitation even though the owner reported a complete sentence.

Curves show saved model predictions, not forced-alignment targets or features measured from the generated WAV. Phoneme-level pitch was trained with interpolation through unvoiced regions. There are no reference test prosody targets for an accuracy comparison.

E3 | English sentence analysis

Original: The rhyth m of the orchestra changed unexpectedly.

Selected checkpoint: 12000 | Conditioning: proxy_086 (ID 86) | Phones: 37 | Waveform duration: 2.833 s

Owner listening: partly noisy; complete. Pitch range: 156.9-235.7 Hz representation. Energy range: 8.40-88.27, raw STFT scale.



The supplied malformed spelling rhyth m is preserved in the original and passed as split tokens. Its saved phones include R AY1 DH EH1 M, not a silently corrected rhythm pronunciation. An early pitch/energy peak does not validate lexical pronunciation. The owner reported complete but partly noisy speech.

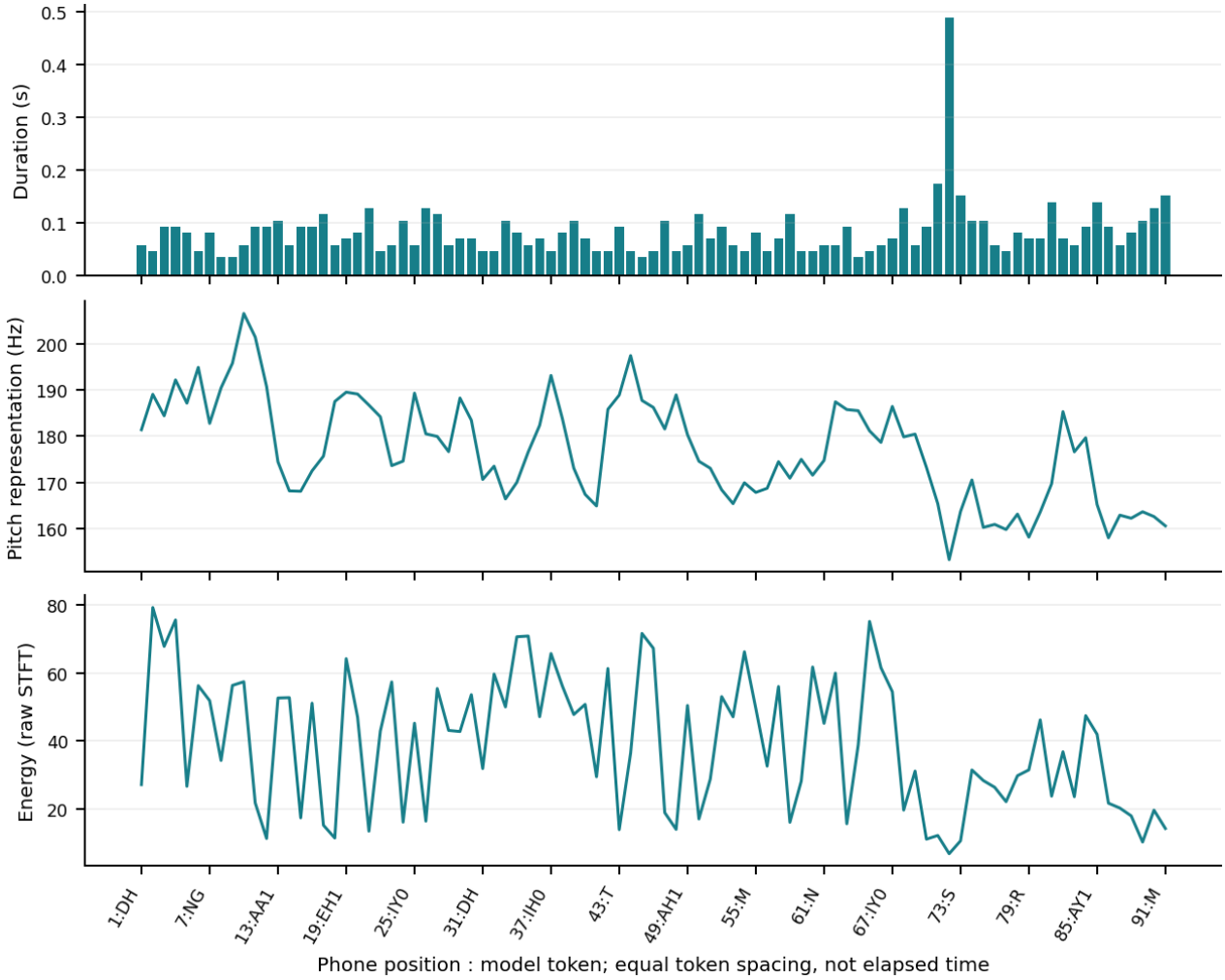
Curves show saved model predictions, not forced-alignment targets or features measured from the generated WAV. Phoneme-level pitch was trained with interpolation through unvoiced regions. There are no reference test prosody targets for an accuracy comparison.

E4 | English sentence analysis

Original: The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time.

Selected checkpoint: 12000 | Conditioning: proxy_086 (ID 86) | Phones: 91 | Waveform duration: 7.500 s

Owner listening: partly noisy; complete. Pitch range: 153.2-206.5 Hz representation. Energy range: 6.70-79.25, raw STFT scale.



Time/date were explicitly expanded to spoken English before tokenization. The comma-associated sp has a predicted duration of about 0.488 s and low energy. Interpolated pitch at this token must not be interpreted as voiced silence. The untranslated andaroo token is retained.

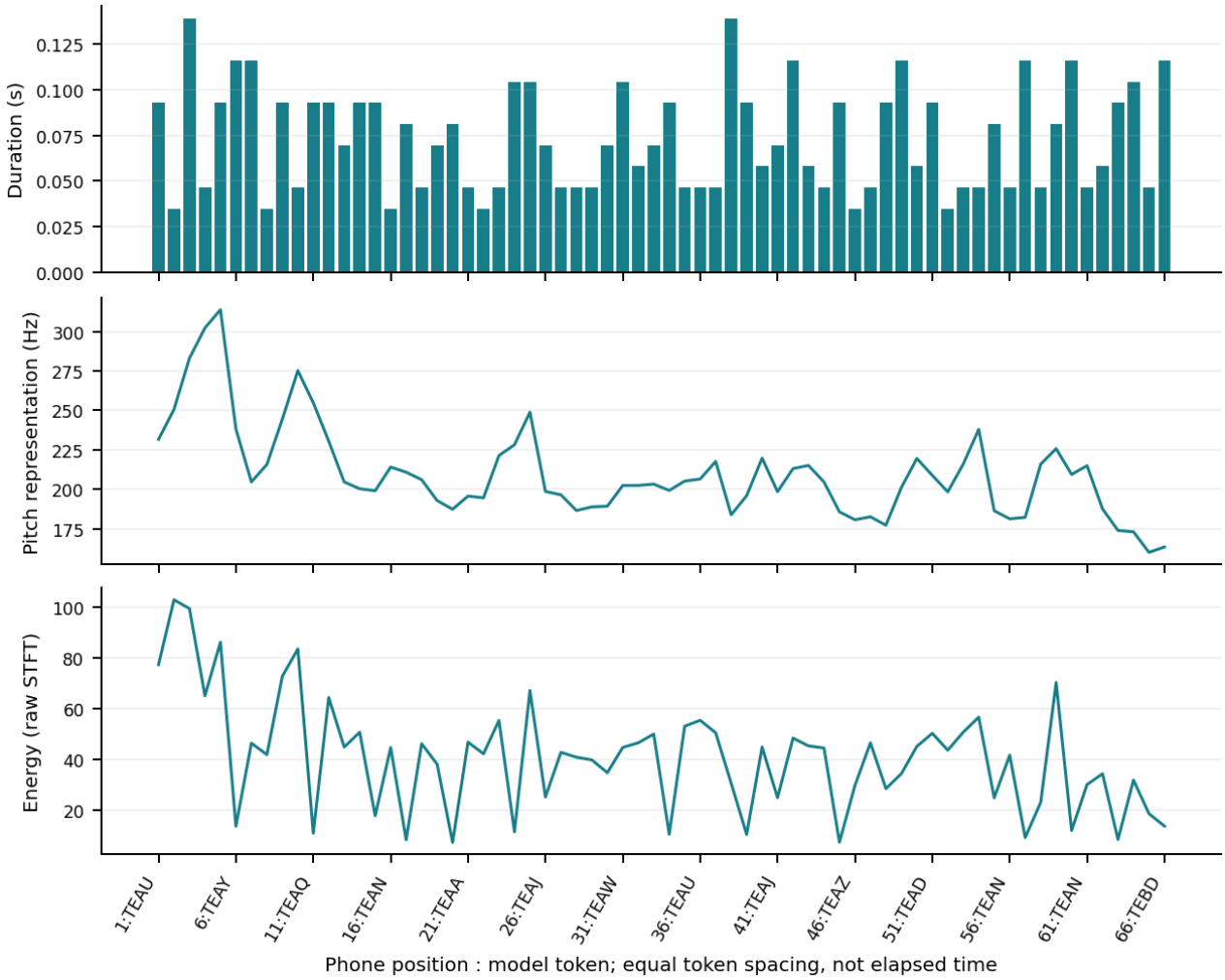
Curves show saved model predictions, not forced-alignment targets or features measured from the generated WAV. Phoneme-level pitch was trained with interpolation through unvoiced regions. There are no reference test prosody targets for an accuracy comparison.

T1 | Telugu sentence analysis

Exact Unicode original and spoken normalization are preserved in Telugu/manifests/test_sentences_tokenized.json. The figure uses fixed model aliases; IPA mapping is supplied in Telugu/lexicon/phone_mapping.json.

Selected checkpoint: 8000 | Conditioning: telugu_female (ID 0) | Phones: 66 | Waveform duration: 4.841 s

Owner listening: clear; complete. Pitch range: 160.2-313.7 Hz representation. Energy range: 7.34-102.93, raw STFT scale.



The specialized term uses eSpeak candidate pronunciation and only training-supported model aliases. Pitch is higher early and generally declines later. Clear/complete listening is a useful subjective observation, but the long term was not independently phoneme-scored.

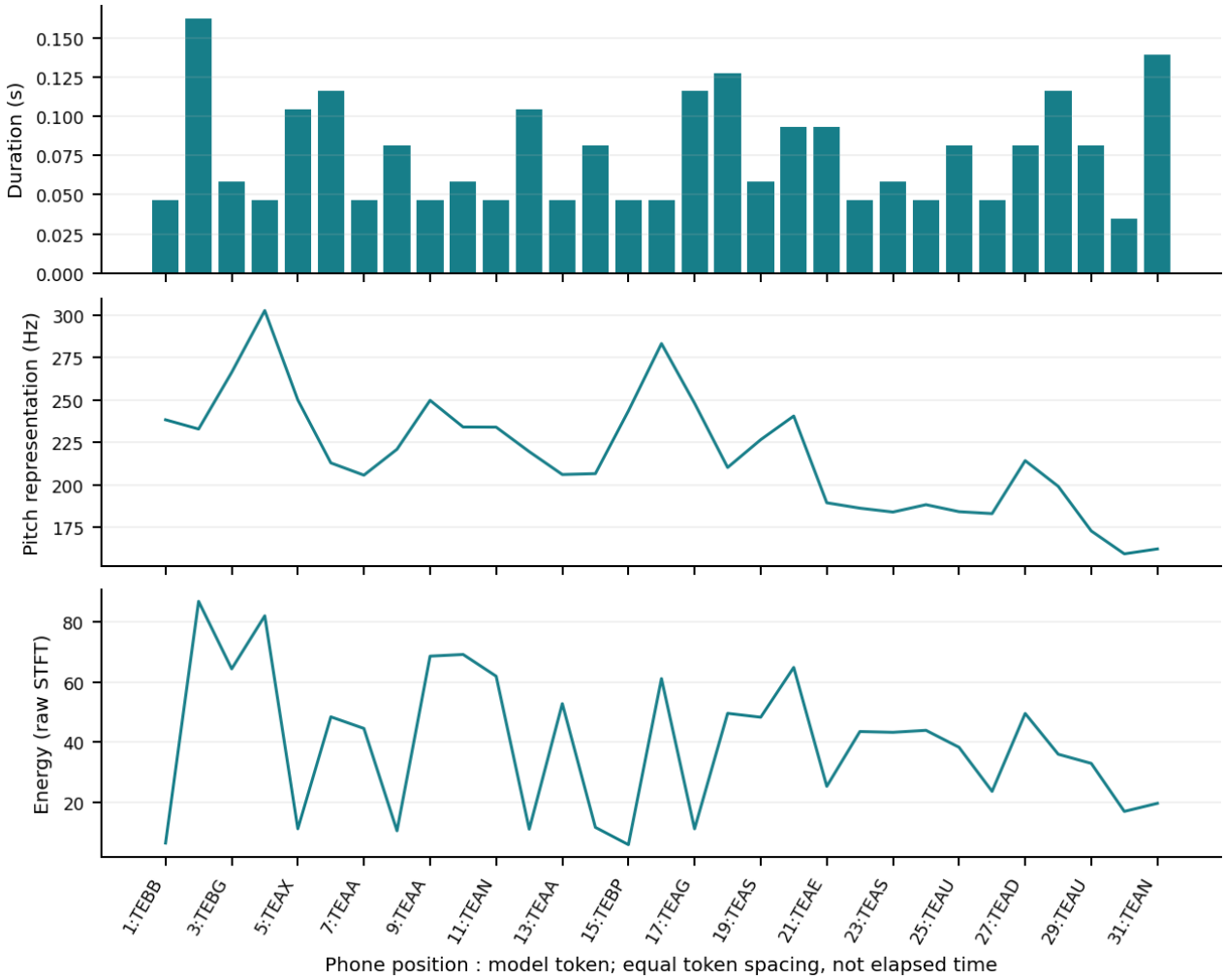
Curves show saved model predictions, not forced-alignment targets or features measured from the generated WAV. Phoneme-level pitch was trained with interpolation through unvoiced regions. There are no reference test prosody targets for an accuracy comparison.

T2a | Telugu sentence analysis

Exact Unicode original and spoken normalization are preserved in Telugu/manifests/test_sentences_tokenized.json. The figure uses fixed model aliases; IPA mapping is supplied in Telugu/lexicon/phone_mapping.json.

Selected checkpoint: 8000 | Conditioning: telugu_female (ID 0) | Phones: 31 | Waveform duration: 2.357 s

Owner listening: clear; complete. Pitch range: 159.4-302.5 Hz representation. Energy range: 5.74-86.83, raw STFT scale.



This sentence uses pattu in a grip context. Longer vowel durations and an overall terminal pitch decline are visible. The owner reported clear and complete speech. No separate word-stress or contrastive meaning-identification score was collected.

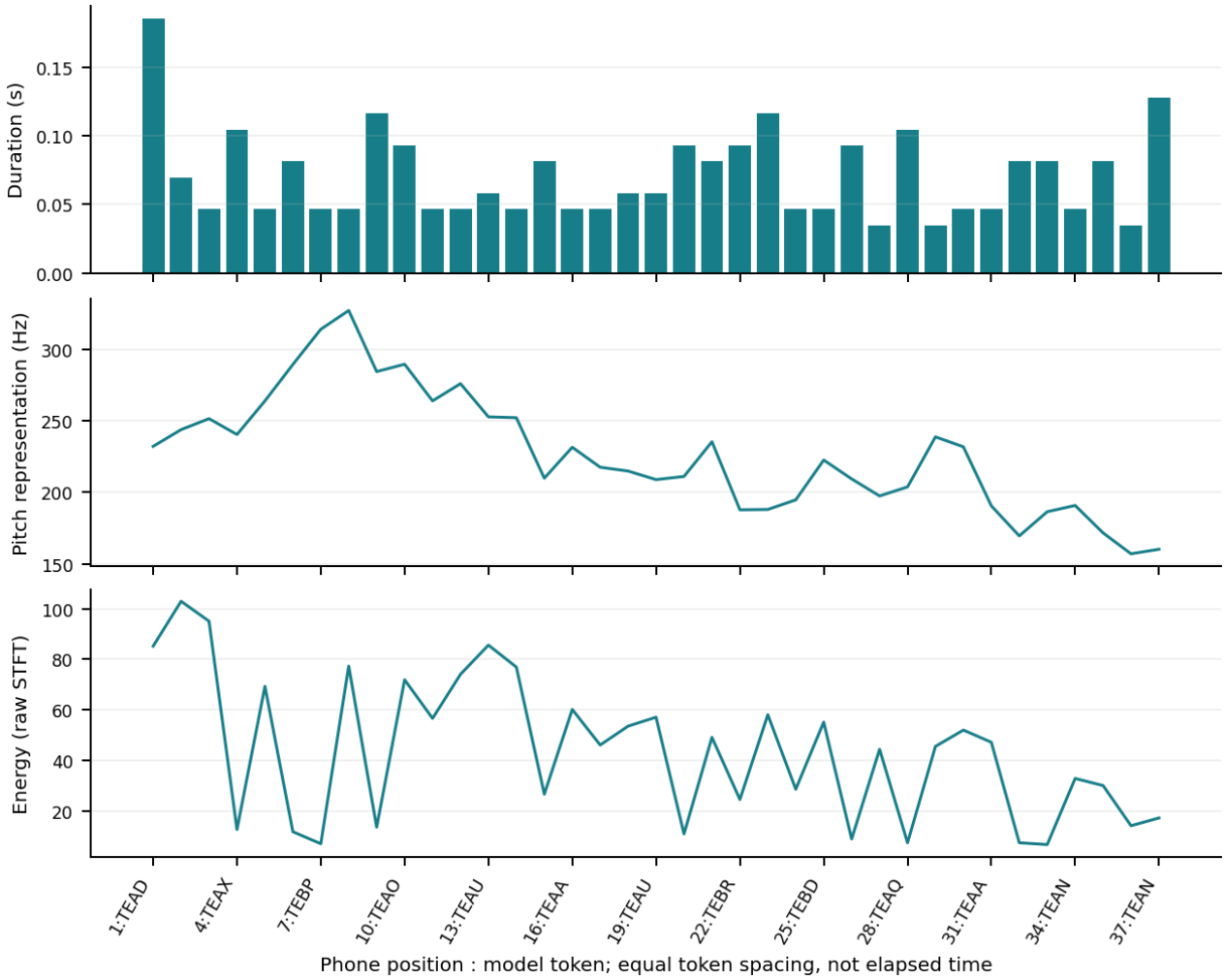
Curves show saved model predictions, not forced-alignment targets or features measured from the generated WAV. Phoneme-level pitch was trained with interpolation through unvoiced regions. There are no reference test prosody targets for an accuracy comparison.

T2b | Telugu sentence analysis

Exact Unicode original and spoken normalization are preserved in Telugu/manifests/test_sentences_tokenized.json. The figure uses fixed model aliases; IPA mapping is supplied in Telugu/lexicon/phone_mapping.json.

Selected checkpoint: 8000 | Conditioning: telugu_female (ID 0) | Phones: 37 | Waveform duration: 2.612 s

Owner listening: clear; complete. Pitch range: 157.0-327.1 Hz representation. Energy range: 6.75-102.88, raw STFT scale.



This sentence uses pattu in the silk-sari context. The initial long vowel is relatively long and the pitch contour has several local peaks. Context meaning is in the text; no special lexical pronunciation override was introduced for pattu.

Curves show saved model predictions, not forced-alignment targets or features measured from the generated WAV. Phoneme-level pitch was trained with interpolation through unvoiced regions. There are no reference test prosody targets for an accuracy comparison.

T3 | Telugu sentence analysis

Exact Unicode original and spoken normalization are preserved in Telugu/manifests/test_sentences_tokenized.json. The figure uses fixed model aliases; IPA mapping is supplied in Telugu/lexicon/phone_mapping.json.

Selected checkpoint: 8000 | Conditioning: telugu_female (ID 0) | Phones: 31 | Waveform duration: 2.043 s

Owner listening: clear; complete. Pitch range: 155.8-316.4 Hz representation. Energy range: 5.34-89.37, raw STFT scale.



The original contains two spaces inside the split word. Normalization collapses whitespace but keeps the resulting two word tokens; it does not automatically rejoin the word or insert a silence phone. Clear/complete listening does not establish a special audible double-space effect.

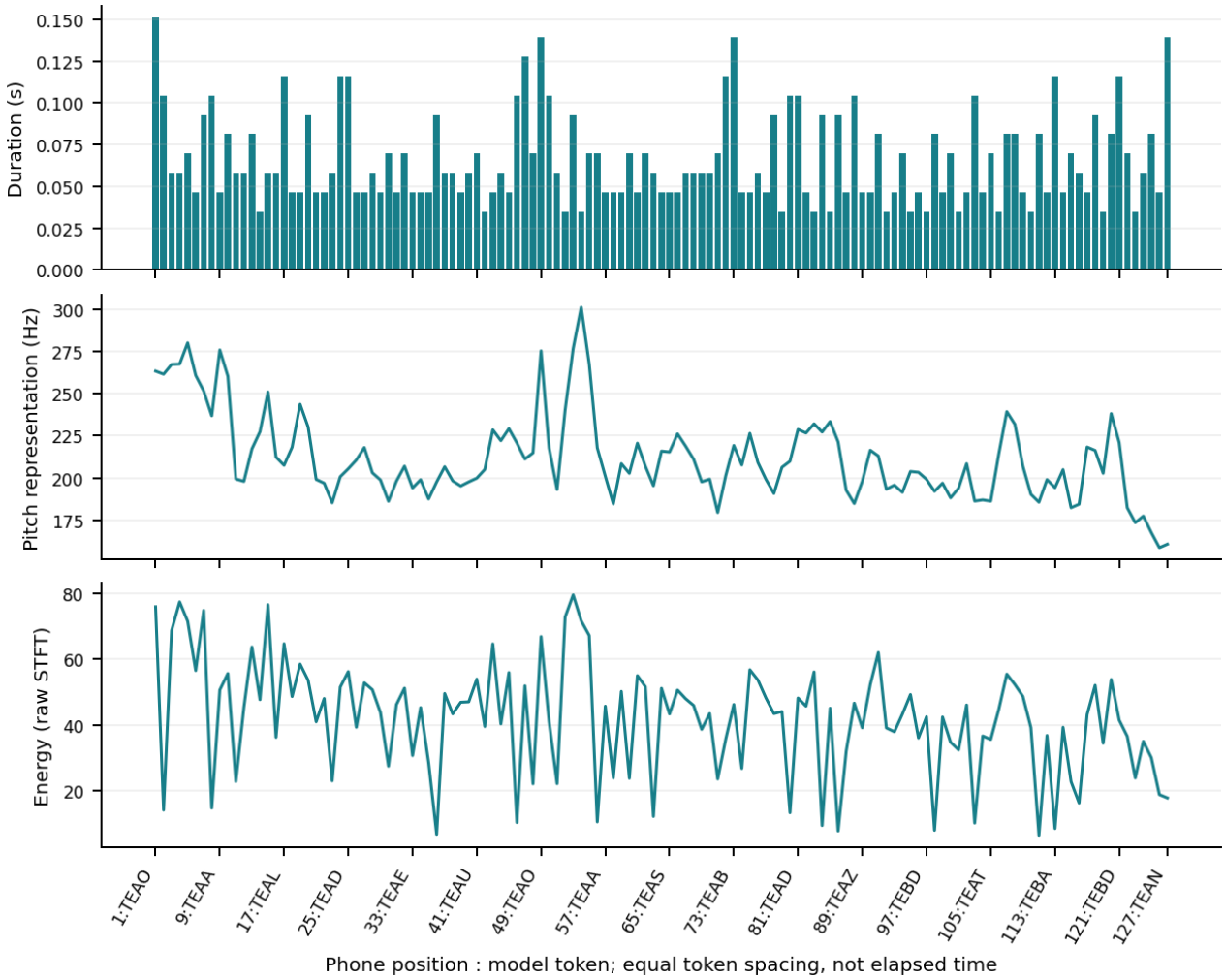
Curves show saved model predictions, not forced-alignment targets or features measured from the generated WAV. Phoneme-level pitch was trained with interpolation through unvoiced regions. There are no reference test prosody targets for an accuracy comparison.

T4 | Telugu sentence analysis

Exact Unicode original and spoken normalization are preserved in Telugu/manifests/test_sentences_tokenized.json. The figure uses fixed model aliases; IPA mapping is supplied in Telugu/lexicon/phone_mapping.json.

Selected checkpoint: 8000 | Conditioning: telugu_female (ID 0) | Phones: 127 | Waveform duration: 8.348 s

Owner listening: clear; complete. Pitch range: 158.5-301.4 Hz representation. Energy range: 6.35-79.53, raw STFT scale.



The year, ordinal date and time were explicitly expanded to Telugu words. With 127 phones, this is the longest Telugu input. The contour has several local peaks and ends lower. Punctuation was normalized to whitespace, so an explicit comma silence token is absent.

Curves show saved model predictions, not forced-alignment targets or features measured from the generated WAV. Phoneme-level pitch was trained with interpolation through unvoiced regions. There are no reference test prosody targets for an accuracy comparison.

6. Limitations and reproduction status

What the evidence establishes

Ten frozen input sequences encode exactly, ten predicted variance sets are numerically consistent, thirty source plots are readable, and normalization metadata matches the saved inverse conversions. Both acoustic models completed their reported training segments and selected checkpoints were verified in Colab. Listening results support clear Telugu examples and partly noisy English examples; they do not establish generalization across arbitrary unseen text, voices or domains.

What remains uncertain

The cause of the English noise is not established. Clear real-mel vocoder reconstruction points toward the predicted acoustic path as a useful investigation area, but it does not rule out vocoder sensitivity to predicted-mel distribution shift. Demographic proxy conditioning, sparse accent coverage per group, alignment quality and mel prediction error are plausible contributors, not proven causes. The small separate repair experiment did not establish a remedy.

Alignment structure is not perceptual boundary accuracy. Pronunciation candidates are not human-verified dictionaries. No sentence-level ASR/WER, MOS or paired listening study was collected. No perceptual improvement percentage can be assigned. Dataset access does not imply redistribution permission; source terms require review before publishing third-party material.

Reproduction and packaging

README.md documents the actual wrapper interfaces, required external files and working-directory requirement for the vocoder. requirements.txt and requirements-preprocessing.txt record observed versions, with CUDA PyTorch and MFA handled separately. A clean installation was not tested in this review. Full raw-to-model reproduction additionally needs the saved split manifests, corpus exports, alignments and historical selection/statistics-fitting cells, which were not uploaded as standalone scripts.

Before publishing: include the final ten WAVs, provide externally accessible selected model/vocoder links, preserve source attribution and licenses, include the missing historical preprocessing resources, and validate inference in a clean environment. Keep final English limitations visible. The uploaded package is documentation and reviewed source, not the complete audio/weight distribution.

Source and artifact references

FastSpeech 2 reference implementation: https://github.com/ming024/FastSpeech2 ; recorded commit d4e79eb52e8b01d24703b2dfc0385544092958f3.

Universal HiFi-GAN source: https://github.com/jik876/hifi-gan ; official pretrained generator identified by SHA-256 in README.md.

Dataset sources: https://huggingface.co/datasets/ai4bharat/Svarah and https://huggingface.co/datasets/SPRINGLab/IndicTTS_Telugu ; pinned revisions listed in Section 1.

Local evidence: Results/*/loss_history.csv; variance_adaptor_predictions/*.{json,npz}; configuration YAMLs; inference_metadata/{stats,speakers}.json; frozen test manifests; reproducibility/selected_checkpoints.json. Source-duration and alignment review statements also use the recorded Colab audit outputs.